PIZZAHUT SALES ANALYSIS

OBJECTIVE

To analysis pizza sales data to provide actionable insights on sales performance, customer performance and revenue generation. By examining sales trends and customer performance, the goal is to identify opportunities to refine marketing strategies, improve operational strategies and ultimately boost revenue.

About the Dataset

This Project contains four dataset Orders, Order_details, Pizzas, Pizza_Types with 48620 sales transaction to analyze. This pizza seles data from January 2015 to December 2015.

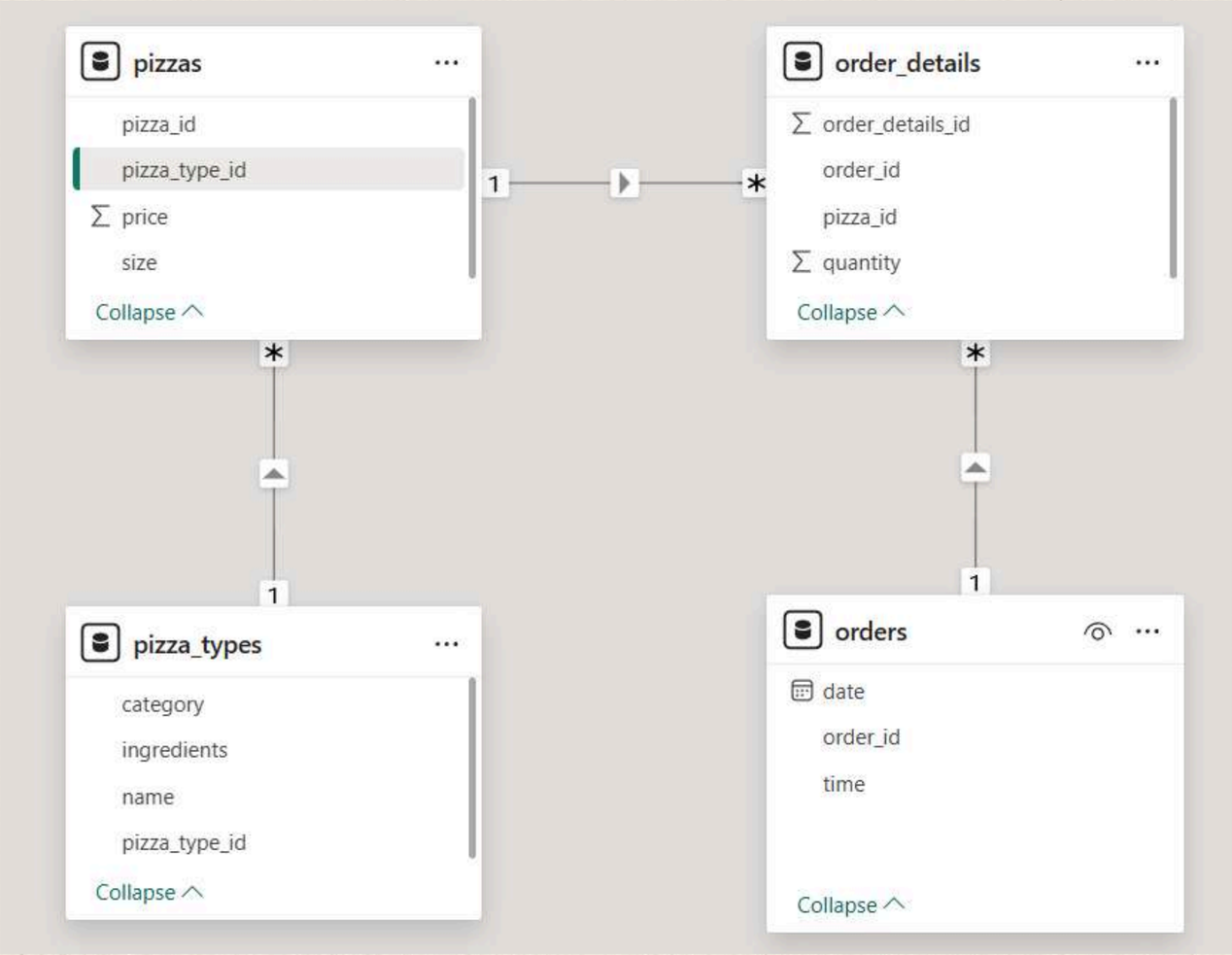
+250
Restaurant

4.9
Good Rate

+100
Pizza menu

INGOUDE COMPANY

DATA MODEL


pizzas
pizza_id
pizza_type_id
price
size
Collapse
order_details
order_details_id
order_id
pizza_id
quantity
Collapse
pizza_types
category
ingredients
name
pizza_type_id
Collapse
orders
date
order_id
time
Collapse
1
*
*
1
*
1

Business Problem

- Retrieve the total number of orders placed.
- Calculate the total revenue generated from pizza sales.
- Identify the highest-priced pizza.
- Identify the most common pizza size ordered.
- List the top 5 most ordered pizza types along with their quantities.
- Join the necessary tables to find the total quantity of each pizza category ordered.
- Determine the distribution of orders by hour of the day.
- Join relevant tables to find the category-wise distribution of pizzas.
- Group the orders by date and calculate the average number of pizzas ordered per day.
- Determine the top 3 most ordered pizza types based on revenue.
- Calculate the percentage contribution of each pizza type to total revenue.
- Analyze the cumulative revenue generated over time.
- Determine the top 3 most ordered pizza types based on revenue for each pizza category.

Q1. Retrieve the total Number of Orders Placed.

```sql
-- Retrieve the total number of orders placed.
select count(order_id) as total_orders from orders;
```

	total_orders
▶	21350

Q2.Calculate the total revenue generated from pizza sales.

```
select round(sum(quantity * price),2) as total_revenue
from pizzas join order_details
on pizzas.pizza_id = order_details.pizza_id;
```

	total_revenue
▶	817860.05

Q3.Identify the highest-priced pizza.

```
select pizza_types.name, pizzas.price
from pizzas
join pizza_types
on pizzas.pizza_type_id = pizza_types.pizza_type_id
order by price desc
limit 1;
```

	name	price
▶	The Greek Pizza	35.95

Q4.Identify the most common pizza size ordered.

```sql
select pizzas.size, count(order_details.order_details_id) as order_count
from pizzas
join order_details
on pizzas.pizza_id = order_details.pizza_id
group by pizzas.size
order by order_count desc;
```

size	order_count
L	18526
M	15385
S	14137
XL	544
XXL	28

Q5.List the top 5 most ordered pizza types along with their quantities.

```sql
select pizza_types.name, sum(order_details.quantity) as quantity
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.name
order by quantity desc
limit 5;
```

name	quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

Q6.Join the necessary tables to find the total quantity of each pizza category ordered.

```
select pizza_types.category,
sum(order_details.quantity) as quantity
from pizza_types
join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category
order by quantity desc;
```

category	quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

Q7.Determine the distribution of orders by hour of the day.

```
select hour(order_time) as hour, count(order_id) as order_count from orders
group by hour(order_time);
```

hour	order_count
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28
10	8
9	1

Q8.Join relevant tables to find the category-wise distribution of pizzas.

```
select category, count(name) from pizza_types
group by category;
```

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5
The Classic Deluxe Pizza	38180.5
The Hawaiian Pizza	32273.25
The Pepperoni Pizza	30161.75
The Spicy Italian Pizza	34831.25
The Italian Supreme Pizza	33476.75
The Sicilian Pizza	30940.5
The Four Cheese Pizza	32265.70000000065
The Mexicana Pizza	26780.75
The Five Cheese Pizza	26066.5

Q9.Group the orders by date and calculate the average number of pizzas ordered per day.

```sql
SELECT
round(AVG(daily_pizzas),0) AS avg_pizzas_per_day
FROM
(
    SELECT
    orders.order_date,
    SUM(order_details.quantity) AS daily_pizzas
    FROM orders
    JOIN order_details
    ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date
) AS daily_orders;
```

avg_pizzas_per_day
138

Q10.Determine the top 3 most ordered pizza types based on revenue.

```
select pizza_types.name,
sum(order_details.quantity * pizzas.price) as revenue
from pizza_types
join pizzas
on pizzas.pizza_type_id = pizza_types.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.name
order by revenue desc limit 3;
```

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

Q11.Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT pizza_types.category,
ROUND(
SUM(order_details.quantity * pizzas.price) /
(
SELECT SUM(order_details.quantity * pizzas.price)
FROM order_details
JOIN pizzas
ON pizzas.pizza_id = order_details.pizza_id
) * 100, 2
) AS revenue
FROM pizza_types
JOIN pizzas
ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN order_details
ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

category	revenue
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

Q12.Analyze the cumulative revenue generated over time.

```
select order_date,
sum(revenue) over(order by order_date) as cum_revenue
from
(select orders.order_date,
sum(order_details.quantity*pizzas.price) as revenue
from order_details
join pizzas
on order_details.pizza_id = pizzas.pizza_id
join orders
on orders.order_id = order_details.order_id
group by orders.order_date) as sales;
```

order_date	cum_revenue
2015-01-01	2713.850000000004
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21526.4
2015-01-10	23990.350000000002
2015-01-11	25862.65
2015-01-12	27781.7
2015-01-13	29831.300000000003
2015-01-14	32358.700000000004

Q13.Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
select name, revenue from
(select category, name, revenue,
rank() over(partition by category order by revenue desc) as rn
from
(select pizza_types.category, pizza_types.name,
sum((order_details.quantity) * pizzas.price) as revenue
from pizza_types
join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as a) as b
where rn <= 3;
```

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5
The Classic Deluxe Pizza	38180.5
The Hawaiian Pizza	32273.25
The Pepperoni Pizza	30161.75
The Spicy Italian Pizza	34831.25
The Italian Supreme Pizza	33476.75
The Sicilian Pizza	30940.5
The Four Cheese Pizza	32265.70000000065
The Mexicana Pizza	26780.75
The Five Cheese Pizza	26066.5

key Insight

- **Most Popular Pizza:** The Classic Deluxe Pizza was the most ordered item across all orders.
- **Top Revenue Generating Pizza:** The Thai Chicken Pizza generated the highest revenue among all types.
- **Best Performing Category:** The Classic category earned the most total revenue across all pizza categories.
- **Peak Order Time:** Most orders were placed between 4:00 PM and 6:00 PM indicating peak customer activity.

key Insight

- **High-value Pizza:** Top prices exceed $35.
- **Popular Sizes:** Most pizzas sold in sizes L and M.
- **Total pizzas sold:** 21350.
- **Total Revenue:** 817860.
- **Average order per Day:** 138.
- **Top-selling categories:**
  - **Classic:** 14888.
  - **Supreme:** 11987.
  - **Veggie:** 11649.
  - **Chicken:** 11050.

Md Aamir Raza

Thank You